8.9 Van't Hoff Equation

This equation gives the quantitative temperature dependence of equilibrium constant (K). The relation between standard free energy change (ΔG°) and equilibrium constant is

$$\Delta G^\circ = -RT\ln K \qquad (1)$$

We know that

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ \qquad (2)$$

Substituting (1) in equation (2)

$$-RT\ln K = \Delta H^\circ - T\Delta S^\circ$$

Rearranging

$$\ln K = \frac{-\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \qquad (3)$$

Differentiating equation (3) with respect to temperature,

$$\frac{d(\ln K)}{dT} = \frac{\Delta H^\circ}{RT^2} \qquad (4)$$

Equation 4 is known as differential form of van't Hoff equation.

On integrating the equation 4, between T_1 and T_2 with their respective equilibrium constants K_1 and K_2.

$$\int_{K_1}^{K_2} d(\ln K) = \frac{\Delta H^\circ}{R}\int_{T_1}^{T_2}\frac{dT}{T^2}$$

$$\left[\ln K\right]_{K_1}^{K_2} = \frac{\Delta H^\circ}{R}\left[-\frac{1}{T}\right]_{T_1}^{T_2}$$

$$\ln K_2 - \ln K_1 = \frac{\Delta H^\circ}{R}\left[-\frac{1}{T_2} + \frac{1}{T_1}\right]$$

$$\ln\frac{K_2}{K_1} = \frac{\Delta H^\circ}{R}\left[\frac{T_2 - T_1}{T_2T_1}\right]$$

$$\log\frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303\,R}\left[\frac{T_2 - T_1}{T_2T_1}\right] \quad \ldots\ldots\ldots (5)$$

Equation 5 is known as integrated form of van't Hoff equation.

Problem:

For an equilibrium reaction K_p = 0.0260 at 25° C ΔH= 32.4 kJmol^{-1}, calculate K_p at 37° C

Solution :

T_1=25 + 273 = 298 K

T_2 = 37 + 273 = 310 K

ΔH = 32.4 KJmol^{-1} = 32400 Jmol^{-1}

R=8.314 JK^{-1} mol^{-1}

K_{p_1} = 0.0260

K_{p_2} = ?

$$\log\frac{K_{p_2}}{K_{p_1}} = \frac{\Delta H^\circ}{2.303\,R}\left[\frac{T_2 - T_1}{T_2T_1}\right]$$

$$\log\frac{K_{p_2}}{K_{p_1}} = \frac{32400}{2.303\times 8.314}\left(\frac{310-298}{310\times 298}\right)$$

$$= \frac{32400\times 12}{2.303\times 8.314\times 310\times 298}$$

$$= 0.2198$$

$$\frac{K_{p_2}}{K_{p_1}} = \text{anti}\log 0.2198 = 1.6588$$

$$K_{p_2} = 1.6588\times 0.026 = 0.0431$$

Evaluate Yourself

5) The equilibrium constant for the following reaction is 0.15 at 298 K and 1 atm pressure.

$$N_2O_4(g) \rightleftharpoons 2NO_2(g);$$

$$\Delta H_f^\circ = 57.32 \text{ KJmol}^{-1}$$

The reaction conditions are altered as follows.

a) The reaction temperature is altered to 100° C keeping the pressure at 1 atm, Calculate the equilibrium constant.

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